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2025 Edition

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The Importance of the Skin

The skin is frequently taken for granted as a passive wrapping around the body, yet it is in fact a highly dynamic, metabolically active organ that serves numerous vital physiological and immunological functions. Every health professional — not only dermatologists — must appreciate the skin's significance, because cutaneous examination can reveal early signs of serious systemic disease, and skin conditions themselves can profoundly alter quality of life equivalent to or worse than that of chronic internal illnesses.

The Body's Largest Organ

The skin is the largest organ of the human body, accounting for approximately 15–20% of total body weight and covering a surface area of roughly 1.5–2.0 m^2 in an average adult. Its sheer size means that it has a substantial metabolic demand, a rich blood supply, and an intimate relationship with virtually every other organ system. Because of this, primary skin disorders and secondary cutaneous manifestations of internal disease are among the most common complaints encountered in all areas of clinical practice — from primary care to emergency medicine and every surgical and medical specialty.

Barrier Function and Homeostasis

The skin provides an essential living biological barrier between the internal milieu and the external environment. This barrier function is multifaceted:

- **Mechanical protection:** The epidermis, particularly the stratum corneum, acts as a physical shield against trauma, abrasion, and penetration by microorganisms, chemicals, and allergens.
- **Chemical protection:** The slightly acidic pH of the skin surface (the "acid mantle," pH approximately 4.5–5.5), produced by sebaceous gland secretions, sweat, and keratinocyte byproducts, inhibits the growth of many pathogenic organisms.
- **Immunological protection:** The skin contains a rich population of immune cells — Langerhans cells (dermal dendritic cells that serve as antigen-presenting cells), dermal macrophages, mast cells, and resident T lymphocytes — collectively termed the skin-associated lymphoid tissue (SALT). These cells provide active immune surveillance and initiate both innate and adaptive immune responses.
- **Microbiome defence:** The commensal skin flora (including *Staphylococcus epidermidis*, *Cutibacterium acnes*, and various fungi) competitively inhibits colonisation by pathogenic organisms.
- **UV protection:** Melanocytes in the basal layer of the epidermis produce melanin, which absorbs and scatters ultraviolet radiation, protecting underlying structures from DNA damage.

Thermoregulation and Fluid Balance

The skin plays a central role in temperature regulation through several mechanisms:

- **Vasodilation and vasoconstriction** of dermal blood vessels, controlled by the sympathetic nervous system, modulate heat loss from the body surface.
- **Eccrine sweat glands** produce sweat, the evaporation of which dissipates heat.
- **Subcutaneous fat** provides thermal insulation.

Additionally, the skin prevents excessive water loss. The stratum corneum, with its lipid-rich intercellular matrix (composed of ceramides, cholesterol, and free fatty acids), forms a waterproof barrier that maintains hydration of the underlying tissues. Disruption of this barrier — as occurs in severe burns, extensive eczema, or toxic epidermal necrolysis — can lead to life-threatening fluid and electrolyte losses.

The Skin as a Window to Systemic Disease

One of the most clinically important aspects of dermatology is that **underlying systemic disease can be heralded by changes on the skin surface**. Numerous internal conditions produce characteristic cutaneous signs that may precede, accompany, or follow the systemic presentation. Examples include:

- **Acanthosis nigricans** (velvety hyperpigmentation of flexural skin) — may indicate insulin resistance, type 2 diabetes mellitus, or an underlying gastrointestinal malignancy (particularly gastric adenocarcinoma).
- **Erythema nodosum** (tender erythematous nodules on the shins) — associated with streptococcal infection, sarcoidosis, inflammatory bowel disease, tuberculosis, and certain drugs.
- **Dermatomyositis** (heliotrope rash, Gottron's papules) — may be a paraneoplastic manifestation, particularly of ovarian, lung, or gastrointestinal cancers.
- **Pyoderma gangrenosum** — associated with inflammatory bowel disease, rheumatoid arthritis, and haematological malignancies.
- **Generalised pruritus without primary skin lesions** — may indicate cholestatic liver disease, chronic kidney disease (uraemic pruritus), haematological malignancy (polycythaemia vera, lymphoma), or thyroid dysfunction.

KEY TEACHING POINTS

Always perform a thorough skin examination as part of the general physical examination — the skin may reveal the first clue to an undiagnosed systemic condition.

Examine the **whole skin**, including the nails, hair, and mucous membranes (oral, genital), not just the area of the patient's presenting complaint.

A validated measure of how much skin disease affects patients' lives is the **Dermatology Life Quality Index (DLQI)**. This is a 10-question questionnaire that assesses the impact of skin disease on symptoms, daily activities, leisure, work/school, personal relationships, and treatment burden over the preceding week. Scores range from 0 (no impact) to 30 (extreme impact). It is widely used in clinical practice and research to quantify disease burden and monitor treatment response.

Psychological and Psychiatric Impact of Skin Disease

Skin diseases can profoundly affect patients' psychiatric health and psychosocial well-being. The visible nature of skin conditions means that patients may experience:

- **Stigma and social isolation** — visible lesions on the face, hands, or other exposed areas can lead to embarrassment, social withdrawal, and avoidance of public situations.
- **Depression and anxiety** — studies consistently show higher rates of clinical depression and generalised anxiety disorder in patients with chronic skin conditions such as psoriasis, eczema, and acne compared with the general population.
- **Body image disturbance** — patients may develop distorted perceptions of their appearance, leading to reduced self-esteem and, in severe cases, body dysmorphic disorder.
- **Sleep disturbance** — pruritus (itching), particularly in conditions like atopic dermatitis, can severely disrupt sleep architecture, leading to fatigue, impaired concentration, and reduced daytime functioning.
- **Impact on relationships and intimacy** — genital or widespread skin disease can impair sexual function and intimate relationships.

- **Occupational impact** — certain professions (e.g., healthcare workers, food handlers, public-facing roles) may be affected by visible skin conditions, leading to workplace discrimination or the need for career modification.

IMPORTANT

Always screen patients with chronic skin disease for psychological distress. Simple screening questions such as "How has your skin condition affected your mood or daily life?" can open the door to further assessment.

The DLQI should be used routinely in dermatology clinics to capture the patient's perspective on disease impact.

Referral to psychology or psychiatry services should be considered when there is evidence of significant psychiatric comorbidity, including suicidal ideation.

Structure and Function of Normal Skin

Understanding the microscopic and macroscopic anatomy of normal skin is fundamental to comprehending how skin diseases develop, how topical therapies penetrate, and how clinical signs manifest.

Layers of the Skin

The skin is composed of three principal layers: the epidermis, the dermis, and the subcutaneous tissue (hypodermis). Each layer has distinct structural components and functions.

The Epidermis

The epidermis is the outermost layer of the skin and is a stratified squamous keratinised epithelium. It is avascular and ranges in thickness from approximately 0.05 mm on the eyelids to over 1.5 mm on the palms and soles. The epidermis is organised into five strata (from deepest to most superficial):

- **Stratum basale (basal layer):** A single layer of columnar to cuboidal keratinocytes resting on the basement membrane. This layer contains stem cells that continuously divide to replenish the epidermis. Melanocytes, which produce melanin, and Merkel cells, which serve as touch receptors, are also located in this layer.
- **Stratum spinosum (prickle cell layer):** Several layers of polyhedral keratinocytes connected by desmosomes (which give the "spiny" appearance on histology). Langerhans cells (dendritic antigen-presenting cells) are most abundant in this layer.
- **Stratum granulosum (granular layer):** Three to five layers of flattened keratinocytes containing keratohyalin granules (which contain profilaggrin and loricrin) and lamellar bodies (Odland bodies) that release lipid contents into the intercellular space, contributing to the epidermal permeability barrier.
- **Stratum lucidum:** A thin, translucent layer found only in thick skin (palms and soles). It consists of densely packed dead keratinocytes filled with eleidin (a transformation product of keratohyalin).
- **Stratum corneum (horny layer):** The outermost layer, consisting of 15–20 layers of dead, flattened, anucleate keratinocytes (corneocytes) embedded in a lipid matrix. This layer is the primary physical and permeability barrier of the skin. Corneocytes are continuously shed (desquamation) and replaced by cells migrating from below.

The process by which keratinocytes differentiate and migrate from the basal layer to the surface is called **keratinisation** (or cornification). The entire epidermal turnover time — from basal cell division to desquamation — takes approximately 28–30 days in normal skin. This turnover rate is accelerated in conditions such as psoriasis (reduced to 3–5 days), leading to the formation of thick, scaly plaques.

The Dermis

The dermis is a connective tissue layer that provides structural support, elasticity, and nourishment to the epidermis. It ranges from 0.3 mm (eyelids) to 3.0 mm (back) in thickness and is divided into two regions:

- **Papillary dermis:** The superficial, thinner layer that interdigitates with the epidermis via dermal papillae. It contains loose areolar connective tissue, capillaries (the subpapillary plexus), nerve endings, and Meissner's corpuscles (light touch receptors).
- **Reticular dermis:** The deeper, thicker layer composed of dense irregular connective tissue with thick bundles of type I collagen fibres and elastic fibres (elastin). It contains larger blood vessels, lymphatic vessels, nerves, and skin appendages (hair follicles, sebaceous glands, eccrine and apocrine sweat glands).

The principal cell type in the dermis is the **fibroblast**, which synthesises collagen, elastin, and ground substance (glycosaminoglycans, particularly hyaluronic acid). Other resident cells include mast cells, macrophages, and dermal dendritic cells.

The Subcutaneous Tissue (Hypodermis)

The subcutaneous tissue lies beneath the dermis and is composed primarily of adipose tissue (fat) organised into lobules separated by fibrous septa containing blood vessels, lymphatics, and nerves. Functions include:

- Thermal insulation
- Mechanical cushioning and shock absorption
- Energy storage
- Hormonal function (adipose tissue is an endocrine organ that produces leptin, adiponectin, and oestrogen via aromatase activity)

Skin Appendages

The skin contains several specialised structures derived from the epidermis that extend into the dermis and subcutaneous tissue:

Hair Follicles

Hair follicles are distributed over the entire skin surface except for the palms, soles, lips, and glans penis. Each hair follicle consists of:

- **Hair bulb:** The deepest portion, containing the dermal papilla (a vascularised connective tissue structure that directs hair growth) and the hair matrix (proliferating keratinocytes that form the hair shaft).
- **Hair shaft:** Composed of three layers — the medulla (innermost, may be absent in fine hair), the cortex (provides strength and pigment), and the cuticle (outermost protective layer of overlapping scales).
- **Inner root sheath (IRS)** and **outer root sheath (ORS)**: Epithelial layers that surround and guide the hair shaft.
- **Arrector pili muscle:** A smooth muscle attached to the follicle that contracts in response to cold or fear, causing the hair to stand erect ("goosebumps").

Hair growth occurs in a cyclical pattern with three phases:

- **Anagen (growth phase):** Lasts 2–6 years for scalp hair. Approximately 85–90% of scalp hairs are in anagen at any given time.
- **Catagen (regression phase):** Lasts 2–3 weeks. The hair follicle shrinks and the lower portion is destroyed.

- **Telogen (resting phase):** Lasts approximately 3 months. The hair is retained in the follicle but is eventually shed. Approximately 10–15% of scalp hairs are in telogen at any given time.

Sebaceous Glands

Sebaceous glands are holocrine glands (the entire cell disintegrates to form the secretion) that are usually associated with hair follicles, forming the pilosebaceous unit. They are most numerous on the face and scalp. Sebaceous glands produce **sebum**, a complex mixture of triglycerides, wax esters, squalene, free fatty acids, and cholesterol. Sebum serves to:

- Lubricate the skin and hair
- Contribute to the acid mantle
- Possess antimicrobial properties

Sebaceous gland activity is primarily regulated by androgens (particularly dihydrotestosterone, DHT), which is why sebaceous gland activity increases at puberty and conditions like acne vulgaris are common in adolescence.

Sweat Glands

There are two main types of sweat glands:

- **Eccrine sweat glands:** Distributed over almost the entire body surface (most numerous on palms, soles, and forehead). They produce a dilute, hypotonic secretion (sweat) primarily composed of water, sodium chloride, urea, and lactate. Eccrine glands are innervated by sympathetic cholinergic fibres and are the primary effectors of thermoregulatory sweating.
- **Apocrine sweat glands:** Found in the axillae, groin, areolae, and perianal region. They produce a viscous, lipid-rich secretion that is initially odorous when metabolised by skin bacteria. Apocrine glands become active at puberty and are innervated by sympathetic adrenergic fibres. Their function in humans is primarily related to scent (pheromone) production.

Nails

The nail unit consists of the nail matrix (the germinal epithelium that produces the nail plate), the nail bed (the epithelium beneath the nail plate that contributes to nail adhesion and growth), the nail plate (the visible hard keratinised structure), the hyponychium (the skin beneath the free edge of the nail), and the proximal and lateral nail folds. Nails protect the distal phalanges, enhance fine manipulation, and serve as a useful clinical indicator of systemic disease.

KEY TEACHING POINTS

The epidermis is avascular — it receives nutrients by diffusion from dermal capillaries across the basement membrane.

Epidermal turnover time is approximately 28–30 days in normal skin but is dramatically accelerated in psoriasis (3–5 days), explaining the thick scale seen clinically.

The stratum corneum is often described as a "bricks and mortar" model — corneocytes (bricks) embedded in a lipid matrix (mortar) of ceramides, cholesterol, and free fatty acids.

Fibroblasts are the key cells of the dermis — they produce collagen (type I and III), elastin, and ground substance. Damage to fibroblasts (e.g., by UV radiation, ageing, or corticosteroid use) leads to skin atrophy and impaired wound healing.

Layers of the Epidermis — Summary

Layer	Key Features	Clinical Relevance
Stratum basale	Single layer of dividing keratinocytes; contains melanocytes and Merkel cells	Origin of most skin cancers (basal cell carcinoma); melanocyte proliferation → naevi and melanoma
Stratum spinosum	Polyhedral keratinocytes with desmosomes; Langerhans cells present	Pemphigus vulgaris — autoantibodies against desmogleins cause acantholysis in this layer
Stratum granulosum	Keratohyalin granules; lamellar body secretion forms lipid barrier	Defects in profilaggrin → ichthyosis vulgaris and atopic dermatitis
Stratum lucidum	Thin, translucent layer; only in thick skin	Provides additional friction and protection on palms and soles
Stratum corneum	Dead, anucleate corneocytes in lipid matrix; primary permeability barrier	Disruption → increased transepidermal water loss; target of topical drug delivery

Skin Examination: A Systematic Approach

A thorough and systematic approach to skin examination is essential for accurate diagnosis. Dermatology is a visual specialty, and the ability to accurately describe skin lesions using standardised terminology is a core clinical skill.

Preparing for Skin Examination

Before examining the patient, ensure the following:

- **Adequate lighting:** Natural daylight is ideal. If unavailable, use a bright, diffuse artificial light source. Poor lighting can lead to missed diagnoses.
- **Full exposure:** Ask the patient to undress to their underwear. Examine the entire skin surface, including the scalp, hair, nails, oral mucosa, and genital skin. Focusing only on the area of the presenting complaint risks missing important incidental findings.
- **Equipment:** Have a dermatoscope (for examining pigmented lesions and skin structures not visible to the naked eye), a magnifying glass, a glass slide for diascopy (blanching test), and a ruler for measuring lesion dimensions.
- **Patient comfort:** Maintain privacy, explain what you are doing, and be sensitive to the patient's comfort and dignity.

The Dermatological History

A structured dermatological history should include:

- **Presenting complaint:** What is the main skin problem? When did it start? How has it changed over time?
- **Symptom assessment:** Is the lesion itchy (pruritic), painful, burning, or asymptomatic? Pruritus is a hallmark of atopic dermatitis, scabies, urticaria, and allergic contact dermatitis.

- **Duration and evolution:** Acute (hours to days), subacute (days to weeks), or chronic (weeks to months)? Is it getting better, worse, or staying the same?
- **Distribution:** Where on the body is the rash? Is it localised, generalised, symmetrical, or asymmetrical? Does it follow a dermatomal pattern (suggesting herpes zoster), or is it in sun-exposed areas (suggesting photosensitivity)?
- **Exacerbating and relieving factors:** Sun exposure, heat, cold, stress, certain foods, medications, contact with specific substances?
- **Previous treatments:** What has the patient already tried? Were they effective? Include over-the-counter products, herbal remedies, and treatments prescribed by other clinicians.
- **Past medical history:** Any history of atopy (eczema, asthma, hay fever), autoimmune disease, immunosuppression, previous skin cancers?
- **Drug history:** A full medication list is essential — many skin conditions are drug-induced. Ask about recent medication changes (within the past 6–8 weeks) and over-the-counter or herbal preparations.
- **Family history:** Atopic conditions, psoriasis, skin cancer, autoimmune diseases?
- **Social history:** Occupation (contact dermatitis risk), hobbies, travel history (tropical infections, sun exposure), alcohol and smoking, sexual history (if relevant to the presentation)?
- **Impact on quality of life:** Use the DLQI or simply ask about the effect on sleep, work, social activities, and mood.

Describing Skin Lesions: The Language of Dermatology

Accurate description of skin lesions using standardised dermatological terminology is essential for communication between clinicians and for forming a differential diagnosis. The key elements to describe are:

Primary Lesions

Primary lesions are the initial pathological changes in the skin that have not been altered by scratching, infection, or treatment:

- **Macule:** A flat, circumscribed area of colour change less than 1 cm in diameter. No elevation or depression of the skin surface. Examples: freckle, flat naevus, petechia.
- **Patch:** A flat, circumscribed area of colour change greater than 1 cm in diameter. Examples: vitiligo patch, port-wine stain, café-au-lait macule.
- **Papule:** A solid, elevated lesion less than 1 cm in diameter. Examples: acne papule, insect bite, lichen planus papule.
- **Plaque:** A solid, elevated, flat-topped lesion greater than 1 cm in diameter, often formed by the coalescence of papules. Examples: psoriasis plaque, lichen planus plaque.
- **Nodule:** A solid, elevated lesion greater than 1 cm in diameter with depth (extends into the dermis or subcutaneous tissue). Examples: dermatofibroma, lipoma, nodular melanoma.
- **Vesicle:** A fluid-filled blister less than 0.5 cm in diameter. Examples: herpes simplex vesicle, dermatitis herpetiformis vesicle.
- **Bulla:** A fluid-filled blister greater than 0.5 cm in diameter. Examples: bullous pemphigoid bulla, friction blister.
- **Pustule:** A pus-filled blister (vesicle containing neutrophils). Examples: acne pustule, folliculitis pustule, pustular psoriasis.

• **Wheal:** A transient, elevated, oedematous plaque caused by dermal oedema. Typically pruritic and blanches on pressure. Examples: urticarial wheal, dermatographism.

• **Cyst:** A closed cavity or sac containing fluid or semi-solid material, lined by epithelium. Examples: epidermoid cyst (sebaceous cyst), milium.

• **Comedo (plural: comedones):** A dilated hair follicle filled with keratin and sebum. Open comedones (blackheads) and closed comedones (whiteheads) are characteristic of acne vulgaris.

Secondary Lesions

Secondary lesions result from modification of primary lesions by scratching, infection, healing, or treatment:

• **Scale (squame):** Flakes of stratum corneum that accumulate on the skin surface. Examples: psoriasis (silvery scale), pityriasis rosea (collarette scale), tinea infections (peripheral scaling).

• **Crust (scab)** Dried serum, blood, or pus on the skin surface. Examples: impetigo (honey-coloured crusts), eczema (serous crusts).

• **Excoriation:** A linear erosion caused by scratching. Examples: prurigo nodularis, atopic dermatitis.

• **Erosion:** A superficial loss of epidermis that does not extend below the dermal-epidermal junction. Heals without scarring. Examples: ruptured vesicles in eczema, superficial burns.

• **Ulcer:** A full-thickness loss of epidermis and dermis (and sometimes subcutaneous tissue). Heals with scarring. Examples: venous ulcer, arterial ulcer, pyoderma gangrenosum, Marjolin's ulcer (squamous cell carcinoma arising in a chronic wound).

• **Fissure:** A linear crack in the skin, often painful. Examples: fissured eczema on the hands, angular cheilitis, athlete's foot (interdigital fissures).

• **Scar:** Fibrous tissue replacement following injury to the dermis. Examples: hypertrophic scar, keloid, atrophic scar (as seen in acne or varicella).

• **Lichenification:** Thickening of the skin with accentuation of skin markings, resulting from chronic rubbing or scratching. Examples: chronic eczema, lichen simplex chronicus.

• **Atrophy:** Thinning of the skin, which may appear wrinkled, translucent, or depressed. Examples: skin atrophy from topical corticosteroids, ageing skin, striae.

Other Descriptive Terms

• **Distribution:** Localised, generalised, widespread, symmetrical, asymmetrical, unilateral, bilateral, dermatomal, flexural, extensor, sun-exposed, acral (hands and feet).

• **Configuration (arrangement):** Discrete, confluent, grouped, linear, annular (ring-shaped), arcuate, serpiginous (snake-like), reticular (net-like), targetoid (target-like).

• **Colour:** Erythematous (red), skin-coloured, hypopigmented, hyperpigmented, violaceous (purple), brown, black, yellow, white.

• **Surface characteristics:** Smooth, rough, warty (verrucous), scaly, crusted, ulcerated, vesicular, pustular.

• **Border:** Well-defined (sharp), ill-defined (blanching), regular, irregular, active edge.

KEY TEACHING POINTS

Always describe skin lesions using standardised terminology — this is the universal language of dermatology and is essential for accurate communication between clinicians.

The distinction between primary and secondary lesions is clinically important: primary lesions suggest the underlying pathological process, while secondary lesions may result from scratching, infection, or treatment.

Diascopy (pressing a glass slide against a lesion) is a simple but invaluable bedside test. Erythematous (inflammatory) lesions blanch because blood is displaced from capillaries. Non-blanching lesions (petechiae, purpura) indicate extravasation of blood into the skin and suggest vasculitis, thrombocytopaenia, or meningococcaemia.

Wood's lamp (ultraviolet light at 365 nm) is useful for detecting certain fungal infections (Microsporum species fluoresce green), bacterial infections (Corynebacterium minutissimum — erythrasma — fluoresces coral pink), and pigmentary disorders (hypopigmented lesions such as vitiligo appear more prominent).

MCQ TRAPS & PITFALLS

A macule is flat — if it is raised, it is a papule or nodule, not a macule.

Vesicles and bullae differ only in size (vesicle < 0.5 cm, bulla > 0.5 cm). Both contain clear fluid; if the fluid is pus, the lesion is a pustule.

A wheal is transient — it typically lasts less than 24 hours in any one location. If a lesion persists for more than 24 hours, consider urticarial vasculitis rather than simple urticaria.

Lichenification is a secondary lesion caused by chronic scratching — it is not a primary diagnosis but a sign of chronic irritation.

Primary Skin Lesions — Quick Reference

Lesion	Size	Description	Examples
Macule	< 1 cm	Flat, colour change only	Freckle, petechia, flat naevus
Patch	> 1 cm	Flat, colour change only	Vitiligo, port-wine stain, café-au-lait macule
Papule	< 1 cm	Solid, elevated	Acne papule, insect bite, seborrhoeic keratosis
Plaque	> 1 cm	Solid, elevated, flat-topped	Psoriasis, lichen planus, eczema plaque
Nodule	> 1 cm	Solid, with depth	Dermatofibroma, lipoma, nodular melanoma
Vesicle	< 0.5 cm	Fluid-filled blister	Herpes simplex, dermatitis herpetiformis, chickenpox
Bulla	> 0.5 cm	Large fluid-filled blister	Bullous pemphigoid, pemphigus vulgaris, friction blister
Pustule	Any	Pus-filled blister	Acne, folliculitis, pustular psoriasis
Wheal	Variable	Transient, oedematous plaque	Urticaria, dermatographism
Cyst	Variable	Epithelium-lined cavity	Epidermoid cyst, milium

Overview of Common Skin Conditions

Eczema (Dermatitis)

Eczema is one of the most common skin conditions encountered in clinical practice. The term "eczema" is often used interchangeably with "dermatitis," though some authorities reserve "dermatitis" for exogenous (externally caused) conditions and "eczema" for endogenous (internally driven) conditions. In practice, the terms are used synonymously.

Eczema is characterised histologically by **spongiosis** (intercellular oedema within the epidermis) and clinically by erythema, vesiculation (in acute lesions), scaling, lichenification (in chronic lesions), and intense pruritus.

Atopic Eczema (Atopic Dermatitis)

Atopic eczema is the most common form of eczema and is part of the **atopic triad** (eczema, asthma, allergic rhinitis/hay fever). It typically begins in infancy or early childhood and may persist into adulthood or resolve with age.

Pathophysiology: Atopic eczema results from a complex interaction of genetic, immunological, and environmental factors. Key mechanisms include:

- **Filaggrin gene mutations:** Loss-of-function mutations in the filaggrin gene (FLG) are the strongest known genetic risk factor for atopic eczema. Filaggrin is essential for epidermal barrier function and hydration. Its deficiency leads to increased transepidermal water loss, dry skin, and enhanced penetration of allergens and irritants.
- **Immune dysregulation:** Acute lesions are dominated by a Th2-type immune response with elevated IL-4, IL-13, and IL-31 (a key pruritogenic cytokine). Chronic lesions show a Th1 and Th22 pattern with increased IFN-γ and IL-22.
- **IgE dysregulation:** Most patients with atopic eczema have elevated serum IgE levels and positive skin prick tests or specific IgE to common environmental allergens (house dust mite, pet dander, pollen, foods).
- **Microbial colonisation:** *Staphylococcus aureus* colonises the skin of over 90% of atopic eczema patients (compared with ~5% of healthy individuals) and can trigger flares through superantigen production and direct inflammation.

Clinical features:

- **Infantile phase (0–2 years):** Erythematous, exudative, crusted lesions on the face (especially the cheeks), scalp, and extensor surfaces of the limbs. The nappy area is often spared.
- **Childhood phase (2–12 years):** Lesions become drier, more lichenified, and localise to flexural sites — the antecubital fossae, popliteal fossae, wrists, and ankles.
- **Adolescent/adult phase:** Flexural lichenification persists. Additional patterns include hand eczema, eyelid eczema, and a "head and neck" distribution. Some patients develop a discoid (nummular) pattern with well-defined coin-shaped plaques.

Diagnostic criteria (UK Working Party): A diagnosis of atopic eczema requires an itchy skin condition (or parental report of scratching/rubbing in a child) plus three or more of the following:

- History of flexural involvement (or cheeks/forehead/extensor surfaces in children under 4 years)
- History of a generally dry skin in the last year
- History of atopic disease in the patient or a first-degree relative (or history of atopic disease in a first-degree relative if the patient is under 4 years)
- Onset under the age of 2 years (not used if the child is under 4 years)

• Visible flexural dermatitis (or dermatitis involving the cheeks/forehead/extensor surfaces in children under 4 years)

Management:

• **Emollients:** The cornerstone of management. Emollients should be applied liberally and frequently (at least twice daily, more during flares). They restore the epidermal barrier, reduce transepidermal water loss, and decrease the need for topical corticosteroids. Ointment-based emollients are generally more effective than creams or lotions (which contain preservatives that may irritate sensitive skin).

• **Topical corticosteroids:** Used to treat flares. The potency should be matched to the severity and site of the eczema. Mild potency (e.g., hydrocortisone 1%) for the face and flexures; moderate potency (e.g., betamethasone valerate 0.025%) for the trunk; potent (e.g., betamethasone valerate 0.1%) for the limbs and body; very potent (e.g., clobetasol propionate 0.05%) for short-term use on recalcitrant areas (palms, soles, lichenified plaques). Use the "fingertip unit" (FTU) to guide the amount of topical steroid to apply — one FTU (the amount of cream/ointment squeezed from the tip of the adult index finger to the distal interphalangeal joint) is sufficient to treat an area equivalent to two adult handprints.

• **Topical calcineurin inhibitors:** Tacrolimus ointment (0.03% and 0.1%) and pimecrolimus cream (1%) are steroid-sparing agents useful for sensitive areas (face, eyelids, flexures) and for long-term maintenance. They do not cause skin atrophy.

• **Wet wrap therapy:** Application of a damp layer of bandage or clothing over emollients and/or topical steroids. Useful for severe flares, particularly in children.

• **Antibiotics:** Used when there is clinical evidence of secondary bacterial infection (increased exudation, crusting, weeping, worsening despite appropriate topical therapy). Flucloxacillin 500mg PO QID x 7 days is first-line for *S. aureus* infection. Consider nasal mupirocin if recurrent infections suggest nasal carriage.

• **Antihistamines:** Sedating antihistamines (e.g., hydroxyzine, chlorpheniramine) may help with sleep disruption due to pruritus but do not directly reduce itch. Non-sedating antihistamines are generally not effective for eczema-associated pruritus.

• **Phototherapy:** Narrowband UVB (NB-UVB) phototherapy is effective for moderate-to-severe eczema refractory to topical therapy.

• **Systemic immunosuppressants:** For severe, refractory disease: ciclosporin, azathioprine, methotrexate, or mycophenolate mofetil.

• **Biologic therapy:** Dupilumab (a monoclonal antibody targeting the IL-4 receptor alpha subunit, thereby blocking IL-4 and IL-13 signalling) is licensed for moderate-to-severe atopic eczema in adults and children aged 6 months and older. It has transformed the management of severe disease.

Contact Dermatitis

Contact dermatitis is an eczematous reaction caused by direct contact with an external substance. It is divided into two main types:

• **Irritant contact dermatitis (ICD):** A non-immunological inflammatory response caused by direct toxic damage to the skin by an irritant substance. It does not require prior sensitisation — anyone can develop ICD if the concentration or duration of exposure is sufficient. Common irritants include water, detergents, solvents, acids, alkalis, and friction. Hand dermatitis is the most common presentation, particularly in occupations involving wet work (healthcare workers, hairdressers, cleaners, food handlers).

• **Allergic contact dermatitis (ACD):** A type IV (delayed-type) hypersensitivity reaction that requires prior sensitisation to the allergen. The patient develops eczema at the site of contact with the allergen, typically 48–72 hours after exposure. Common allergens include nickel (jewellery, belt buckles), chromium (leather, cement), fragrances (cosmetics, toiletries), preservatives (methylisothiazolinone in wet wipes and cosmetics), rubber accelerators (gloves, shoes), and plants (poison ivy, primula).

Diagnosis of ACD: Patch testing is the gold standard diagnostic test. Standardised allergen series are applied to the patient's back in aluminium Finn chambers and left in place for 48 hours. The patches are removed and the test is read at 48 hours and again at 96–120 hours. A positive reaction (erythema, papules, vesicles at the test site) confirms sensitisation to that allergen.

Management: Identification and avoidance of the causative agent is the most important intervention. Topical corticosteroids and emollients are used to treat active flares. For severe or widespread ACD, a short course of oral prednisolone (e.g., 30–40mg daily for 5–7 days, then tapered) may be necessary.

Seborrhoeic Dermatitis

Seborrhoeic dermatitis is a common, chronic, relapsing eczematous condition that affects sebum-rich areas of the skin (scalp, face — particularly the nasolabial folds, eyebrows, and ears — and upper chest). It is associated with the yeast *Malassezia furfur* (formerly *Pityrosporum ovale*), which is a normal commensal of the skin but triggers an inflammatory response in susceptible individuals.

Clinical features: Erythematous patches with greasy, yellowish scale. In infants, it presents as "cradle cap" (thick, greasy scale on the scalp). In adults, it may present as dandruff (scalp scaling without significant erythema) or as more extensive facial and truncal involvement. It is more severe and more common in patients with HIV/AIDS and Parkinson's disease.

Management: Antifungal agents (ketoconazole shampoo or cream, ciclopirox olamine) are first-line. Low-potency topical corticosteroids (hydrocortisone 1%) can be used for short courses to control inflammation. Calcineurin inhibitors (tacrolimus, pimecrolimus) are useful for facial disease to avoid steroid side effects.

Discoid (Nummular) Eczema

Discoid eczema presents as well-defined, coin-shaped (round or oval), erythematous, crusted, and vesicular plaques, typically on the limbs. It is more common in middle-aged and elderly men and is often very pruritic. The aetiology is unclear but may be related to dry skin, irritant exposure, or localised infection. Management includes emollients, topical corticosteroids (moderate to potent), and treatment of any secondary infection.

Varicose (Stasis) Eczema

Varicose eczema occurs on the lower legs in patients with chronic venous insufficiency. It presents as erythematous, scaly, sometimes exudative patches around the medial malleolus, often accompanied by other signs of venous disease (varicose veins, venous ulcers, lipodermatosclerosis, haemosiderin staining). Management includes emollients, topical corticosteroids, and compression therapy to address the underlying venous hypertension.

KEY TEACHING POINTS

Emollients are the foundation of eczema management — they should be used continuously, even when the skin is clear, to prevent flares.

The "fingertip unit" (FTU) is a practical guide for patients: one FTU covers an area equal to two adult handprints. For a full-body application in an adult, approximately 20–30 g of emollient is needed per application.

Patch testing should be considered in any patient with eczema that is localised, atypical, or refractory to standard treatment — it may reveal a contact allergen that can be avoided.

Atopic eczema is associated with the "atopic march" — the natural progression from eczema in infancy to food allergy, allergic rhinitis, and asthma in later childhood. Not all atopic children follow this progression.

OSCE PEARLS

When examining a patient with eczema, always check the hands, feet, and nails — hand involvement significantly impacts function and quality of life.

Look for signs of secondary infection: increased exudation, honey-coloured crusting, pustules, or worsening despite appropriate treatment.

In a child with eczema, ask about the impact on sleep and school performance — this information is important for assessing disease severity and guiding treatment decisions.

Always ask about the patient's current use of emollients and topical steroids — many patients underuse emollients and are fearful of topical steroids ("steroid phobia"), which can lead to poorly controlled disease.

Types of Eczema — Comparison

Type	Key Features	Typical Sites	Key Management
Atopic eczema	Itchy, erythematous, dry, lichenified; associated with atopy	Flexures (antecubital, popliteal), face (infants)	Emollients, topical steroids, calcineurin inhibitors, dupilumab
Irritant contact dermatitis	Non-immunological; caused by direct toxic damage	Hands (wet work occupations)	Avoid irritants, emollients, topical steroids
Allergic contact dermatitis	Type IV hypersensitivity; requires patch testing	Site of allergen contact (e.g., earlobes — nickel)	Identify and avoid allergen, patch testing, topical steroids
Seborrhoeic dermatitis	Greasy scale in sebum-rich areas; associated with Malassezia	Scalp, face (nasolabial folds, eyebrows), chest	Antifungals (ketoconazole), mild topical steroids
Discoid eczema	Coin-shaped plaques; often very itchy	Limbs (especially lower legs)	Emollients, moderate-potent topical steroids
Varicose eczema	Associated with venous insufficiency	Lower legs (medial malleolus)	Emollients, topical steroids, compression therapy

Psoriasis

Psoriasis is a chronic, immune-mediated inflammatory skin disease affecting approximately 2–3% of the global population. It is characterised by well-demarcated, erythematous, scaly plaques and is associated with several significant comorbidities, including psoriatic arthritis, cardiovascular disease, metabolic syndrome, and depression.

Pathophysiology

Psoriasis is driven by a complex interplay of genetic susceptibility and immune dysregulation. The key immunological pathway involves:

- **Th17/IL-23 axis:** Dendritic cells in the skin produce IL-23, which activates Th17 cells to produce IL-17A, IL-17F, and IL-22. These cytokines drive keratinocyte hyperproliferation (accelerated epidermal turnover from ~28 days to 3–5 days), neutrophil recruitment, and the formation of the characteristic psoriatic plaque.

- **Genetic factors:** HLA-Cw6 (MHC class I allele) is the strongest genetic association. Other susceptibility genes involve immune regulation (IL12B, IL23R, TNFAIP3) and epidermal barrier function (LCE gene cluster).
- **Environmental triggers:** Streptococcal pharyngitis (particularly in guttate psoriasis), trauma (Koebner phenomenon — new psoriatic lesions developing at sites of skin injury), drugs (lithium, beta-blockers, antimalarials, rapid withdrawal of systemic corticosteroids), stress, smoking, and alcohol.

Clinical Features

Chronic Plaque Psoriasis (Psoriasis Vulgaris)

This is the most common form, accounting for approximately 85–90% of cases. It presents as:

- Well-demarcated, erythematous plaques with silvery-white scale
- Symmetrical distribution, typically affecting the extensor surfaces (elbows, knees), scalp, sacrum, and lower back
- The scale is thick and lamellar — when scraped, it reveals pinpoint bleeding points (Auspitz sign), which result from the dilated, tortuous capillaries in the dermal papillae being traumatised
- Nail involvement is common (up to 50% of patients): pitting (small depressions in the nail plate), onycholysis (separation of the nail plate from the nail bed), subungual hyperkeratosis, and oil-drop (salmon-patch) discolouration

Guttate Psoriasis

Guttate psoriasis presents as numerous small (1–10 mm), drop-shaped, erythematous, scaly papules and plaques, typically distributed over the trunk and proximal limbs. It is most common in children and young adults and is frequently triggered by group A beta-haemolytic streptococcal pharyngitis. It may resolve spontaneously within a few months or progress to chronic plaque psoriasis.

Pustular Psoriasis

Pustular psoriasis is characterised by sterile pustules (collections of neutrophils) on an erythematous base. It may be generalised (von Zumbusch type — a dermatological emergency with fever, malaise, and widespread pustulation) or localised (palmoplantar pustulosis — pustules on the palms and soles).

Erythrodermic Psoriasis

Erythroderma (involvement of >90% of the body surface area with erythema and scaling) is a rare but life-threatening complication of psoriasis. It can be triggered by severe sunburn, infection, drugs, or abrupt withdrawal of systemic corticosteroids. Patients are at risk of high-output cardiac failure, hypothermia, fluid and electrolyte imbalance, and secondary infection. Hospitalisation is required.

Assessment of Severity

The severity of psoriasis is assessed using several validated tools:

- **PASI (Psoriasis Area and Severity Index):** A composite score (0–72) that assesses the extent of body surface area involvement and the severity of erythema, induration (thickness), and scaling in four body regions. Used primarily in clinical trials.
- **BSA (Body Surface Area):** The percentage of body surface area affected by psoriasis. One palm (including fingers) represents approximately 1% BSA.
- **DLQI (Dermatology Life Quality Index):** Assesses the impact of psoriasis on quality of life (as described earlier).

• **Static Physician's Global Assessment (sPGA):** A physician-rated global assessment of overall disease severity.

Severity classification (NICE guidelines):

• **Mild:** PASI < 10, DLQI ≤ 10, BSA < 10%

• **Moderate-to-severe:** PASI ≥ 10, DLQI > 10, BSA ≥ 10%, or involvement of high-impact sites (face, palms, soles, genitalia), or significant functional impairment

Management

Topical Therapy (First-Line for Mild Disease)

• **Emollients:** Reduce scaling and pruritus, improve the penetration of other topical agents.

• **Topical corticosteroids:** Used for flares, particularly on the face, flexures, and genitalia. Potent or very potent steroids may be used for short courses on the body. The vitamin D analogue/steroid combination (calcipotriol/betamethasone dipropionate — Enstilar foam or Daivobet ointment) is a highly effective first-line option for body plaques.

• **Vitamin D analogues:** Calcipotriol (Dovonex) and tacalcitol. They normalise keratinocyte proliferation and differentiation. Safe for long-term use. Should not exceed recommended weekly quantities (calcipotriol: 100 g/week for adults) to avoid hypercalcaemia.

• **Coal tar:** Reduces scaling and inflammation. Available in shampoos, creams, and bath additives. Effective but messy and malodorous, which limits patient acceptability.

• **Dithranol (anthralin):** Effective for stable plaque psoriasis. Applied for short contact (30 minutes, then washed off). Stains skin, clothing, and bathroom surfaces purple-brown.

• **Topical retinoids:** Tazarotene 0.05–0.1% gel. Effective but can cause irritation. Contraindicated in pregnancy.

Phototherapy

• **Narrowband UVB (NB-UVB, 311 nm):** The most commonly used phototherapy modality. Administered 2–3 times per week in a phototherapy cabinet. Effective for moderate-to-severe psoriasis. The main side effects are erythema (sunburn) and, with long-term use, increased skin cancer risk.

• **PUVA (Psoralen + UVA):** Oral or topical psoralen (a photosensitising agent) followed by UVA exposure. More effective than NB-UVB but with a higher long-term risk of skin cancer (particularly squamous cell carcinoma). Now largely superseded by NB-UVB.

Systemic Therapy (for Moderate-to-Severe Disease)

• **Methotrexate:** 7.5–25 mg once weekly (oral or subcutaneous), with folic acid 5 mg weekly (taken 24–48 hours after methotrexate to reduce side effects). Hepatotoxicity is the main long-term concern — liver biopsy or FibroScan monitoring is recommended. Contraindicated in pregnancy (teratogenic).

• **Ciclosporin:** 2.5–5 mg/kg/day in two divided doses. Rapid onset of action, useful for acute flares and as a "bridge" therapy. Nephrotoxicity and hypertension are the main concerns — limit use to 1–2 years.

• **Acitretin:** A retinoid (25–50 mg daily). Effective for pustular and erythrodermic psoriasis. Teratogenic — women of childbearing potential must use effective contraception during treatment and for 3 years after stopping acitretin.

• **Dimethyl fumarate (Skilarence):** An immunomodulator licensed for moderate-to-severe plaque psoriasis. GI side effects (diarrhoea, abdominal pain, flushing) are common initially but often improve with time.

• **Apremilast:** A phosphodiesterase-4 (PDE4) inhibitor. Moderate efficacy, oral administration. Side effects include diarrhoea, nausea, and weight loss.

Biologic Therapy

Biologic therapies target specific components of the immune system and have revolutionised the treatment of moderate-to-severe psoriasis:

• **Anti-TNF agents:** Adalimumab, etanercept, infliximab. Effective for psoriasis and psoriatic arthritis.

• **Anti-IL-12/23 agent:** Ustekinumab (targets the p40 subunit shared by IL-12 and IL-23).

• **Anti-IL-23 agents:** Guselkumab, risankizumab, tildrakizumab (target the p19 subunit of IL-23). High efficacy and durable responses.

• **Anti-IL-17 agents:** Secukinumab, ixekizumab (target IL-17A); brodalumab (targets the IL-17 receptor). Very high efficacy — PASI 90 response rates of 65–75% at week 12.

KEY TEACHING POINTS

Psoriasis is a systemic inflammatory disease, not just a skin condition. Patients have an increased risk of cardiovascular disease, metabolic syndrome (obesity, type 2 diabetes, dyslipidaemia, hypertension), psoriatic arthritis, inflammatory bowel disease, and depression. Screen for these comorbidities regularly.

The Koebner phenomenon (isomorphic response) — new psoriatic lesions developing at sites of skin trauma (scratching, surgical wounds, sunburn) — is characteristic of psoriasis and occurs in approximately 25% of patients.

Nail psoriasis is often overlooked but can cause significant functional impairment and pain. It is also a risk factor for psoriatic arthritis.

Before starting systemic immunosuppressive therapy (methotrexate, ciclosporin, biologics), screen for latent tuberculosis (chest X-ray, interferon-gamma release assay), hepatitis B/C, and HIV. Ensure vaccinations are up to date (avoid live vaccines in immunosuppressed patients).

MCQ TRAPS & PITFALLS

Guttate psoriasis is triggered by streptococcal infection — throat swab and ASO titres should be checked.

Erythrodermic psoriasis is a medical emergency — do NOT treat with systemic corticosteroids, as withdrawal can trigger pustular psoriasis.

Methotrexate is given ONCE WEEKLY, not daily. Daily dosing is a common prescribing error that can cause fatal bone marrow suppression.

Acitretin is teratogenic and requires contraception for 3 years after stopping — this is longer than any other systemic psoriasis therapy.

Biologic Therapies for Psoriasis — Overview

Biologic	Target	Dosing	Key Considerations
Adalimumab	TNF-α	80 mg SC loading, then 40 mg every 2 weeks	Screen for TB and hepatitis B/C before starting
Ustekinumab	IL-12/23 (p40)	45 mg or 90 mg SC at weeks 0, 4, then every 12 weeks	Weight-based dosing

Biologic	Target	Dosing	Key Considerations
Secukinumab	IL-17A	300 mg SC weekly for 5 weeks, then monthly	Monitor for candidiasis and IBD exacerbation
Guselkumab	IL-23 (p19)	100 mg SC at weeks 0, 4, then every 8 weeks	High efficacy; convenient dosing interval
Risankizumab	IL-23 (p19)	150 mg SC at weeks 0, 4, then every 12 weeks	Among the highest PASI 90 response rates

Acne Vulgaris

Acne vulgaris is a chronic inflammatory disease of the pilosebaceous unit that primarily affects adolescents and young adults. It is one of the most common skin conditions worldwide and can have a significant psychological impact.

Pathophysiology

Four key factors contribute to the development of acne:

- **Excess sebum production:** Androgens (particularly DHT) stimulate sebaceous gland hypertrophy and increased sebum production. This is why acne typically begins at puberty.
- **Follicular hyperkeratinisation:** Abnormal desquamation of keratinocytes within the follicle leads to the formation of a microcomedo (the precursor lesion of all acne lesions). The microcomedo may be open (blackhead — open comedo) or closed (whitehead — closed comedo).
- **Colonisation with Cutibacterium acnes (C. acnes):** This anaerobic bacterium colonises the pilosebaceous follicle and triggers inflammation through activation of Toll-like receptors (TLR-2, TLR-4), production of pro-inflammatory cytokines (IL-1β, TNF-α), and complement activation.
- **Inflammation:** Inflammation is now recognised as a key driver of acne from the earliest stages, not just a consequence of C. acnes colonisation.

Clinical Features

Acne typically affects the face (99% of cases), back (60%), and chest (15%). Lesions include:

- **Non-inflammatory lesions:** Open comedones (blackheads) and closed comedones (whiteheads)
- **Inflammatory lesions:** Papules, pustules, nodules, and cysts
- **Sequelae:** Post-inflammatory hyperpigmentation (PIH), post-inflammatory erythema (PIE), and scarring (ice-pick, boxcar, rolling, hypertrophic, or keloid scars)

Management

Treatment is guided by the severity and type of acne:

Mild Acne (Predominantly Comedonal)

- **Topical retinoids:** Adapalene 0.1% cream/gel (first-generation retinoid, available over the counter) or tretinoin 0.025–0.05% cream. They normalise follicular keratinisation and have anti-inflammatory properties. Applied at night (retinoids are degraded by UV light). May cause initial irritation — start with alternate-night application and increase to nightly as tolerated. Teratogenic — contraindicated in pregnancy.

• **Benzoyl peroxide (BPO):** 2.5–10% cream/gel. Antimicrobial (kills C. acnes without resistance) and comedolytic. Applied once or twice daily. Can bleach clothing and bedding. Often combined with a topical antibiotic to reduce antibiotic resistance.

Mild-to-Moderate Acne (Papulopustular)

• **Combination therapy:** BPO + topical antibiotic (clindamycin 1% or erythromycin 2%) — available as fixed-dose combinations (Duac, Epiduo). BPO prevents the development of antibiotic resistance.

• **Topical retinoid + BPO:** Available as fixed-dose combinations (Epiduo — adapalene 0.1%/BPO 2.5%).

Moderate-to-Severe Acne (Nodulocystic or Scarring)

• **Oral antibiotics:** Lymecycline 408 mg once daily or doxycycline 100 mg once daily (first-line). Tetracyclines have anti-inflammatory as well as antimicrobial properties. Treatment duration should be limited to 3–6 months to minimise antibiotic resistance. Always co-prescribe a topical retinoid or BPO to maintain remission after stopping the antibiotic. Tetracyclines are contraindicated in pregnancy and children under 12 years (risk of tooth discolouration and impaired bone growth).

• **Combined oral contraceptive pill (COCP):** Useful in women with acne, particularly if there is evidence of androgen excess (hirsutism, irregular periods). Co-cyprindiol (Dianette/Cyproterone acetate + ethinylestradiol) is licensed for acne but should be used cautiously due to an increased risk of venous thromboembolism.

• **Isotretinoin (13-cis-retinoic acid):** The most effective treatment for severe, scarring, or treatment-resistant acne. Dose: 0.5–1.0 mg/kg/day for 16–24 weeks (target cumulative dose: 120–150 mg/kg). Isotretinoin reduces sebum production by up to 90%, normalises follicular keratinisation, reduces C. acnes colonisation, and has anti-inflammatory effects. Side effects are dose-dependent and include: cheilitis (dry, cracked lips — universal), dry skin and mucous membranes, photosensitivity, myalgias, hypertriglyceridaemia, and transient elevation of liver transaminases. **Teratogenicity is the most serious concern** — isotretinoin is absolutely contraindicated in pregnancy. Women of childbearing potential must use effective contraception (two forms, or abstinence) for 1 month before, during, and for 1 month after treatment. In the UK, isotretinoin can only be prescribed by or under the supervision of a dermatologist. Monitoring includes baseline and monthly blood tests (liver function, lipids) and pregnancy testing.

KEY TEACHING POINTS

Acne is not caused by poor hygiene — excessive washing can worsen irritation. Gentle cleansing twice daily is sufficient.

Dietary factors: High glycaemic index foods and dairy (particularly skim milk) have been associated with worsening acne in some studies, though the evidence is not strong enough to make firm dietary recommendations.

Post-inflammatory hyperpigmentation (PIH) is more common in patients with darker skin tones (Fitzpatrick skin types IV–VI) and can be more distressing than the acne itself. Treatment includes sun protection, topical retinoids, azelaic acid, hydroquinone (short-term), and chemical peels.

Isotretinoin does NOT cause depression or suicidal ideation — large-scale studies have not confirmed this association. However, acne itself is associated with significant psychological distress, and many patients report improvement in mood and quality of life after successful isotretinoin treatment.

Acne Treatment Algorithm

Severity	First-Line Treatment	Second-Line / Escalation
Mild (comedonal)	Topical retinoid (adapalene) or BPO	Combination topical therapy
Mild-moderate (papulopustular)	Topical retinoid + BPO ± topical antibiotic	Oral antibiotic (lymecycline/doxycycline) + topical retinoid/BPO
Moderate-severe (nodulocystic)	Oral antibiotic + topical retinoid/BPO	Isotretinoin (dermatologist-supervised)
Hormonal acne (women)	COCP ± topical therapy	Spironolactone (off-label), isotretinoin

Skin Infections

Bacterial Infections

Impetigo

Impetigo is a highly contagious superficial bacterial skin infection, most common in preschool children. It is caused by *Staphylococcus aureus* (most common) or *Streptococcus pyogenes* (group A streptococcus).

- **Non-bullous impetigo (70% of cases):** Begins as erythematous macules that rapidly progress to vesicles or pustules, which rupture to form characteristic honey-coloured crusts. Typically affects the face (around the nose and mouth) and limbs. Local lymphadenopathy may be present.
- **Bullous impetigo (30% of cases):** Caused by *S. aureus* producing exfoliative toxins (ETA, ETB). Presents as flaccid, fragile bullae that rupture easily, leaving a thin, brown, "varnish-like" crust. More common in neonates and young children.

Management: Topical fusidic acid 2% cream/ointment TID x 5 days is first-line for localised disease. Topical mupirocin 2% is an alternative. Oral flucloxacillin 500 mg QID x 7 days (or erythromycin if penicillin-allergic) is used for extensive or systemic disease. Nasal mupirocin may be considered for recurrent cases to eradicate nasal carriage.

Cellulitis and Erysipelas

Cellulitis and erysipelas are acute bacterial infections of the dermis and subcutaneous tissue (cellulitis) or the upper dermis and superficial lymphatics (erysipelas). They are most commonly caused by *Streptococcus pyogenes* (group A streptococcus) and *Staphylococcus aureus*.

- **Erysipelas:** Presents as a well-demarcated, raised, erythematous, oedematous plaque with a sharp, advancing border. Typically affects the face (butterfly distribution) or lower legs. Associated with fever, malaise, and systemic toxicity.
- **Cellulitis:** Presents as a poorly demarcated area of erythema, warmth, swelling, and tenderness. The borders are diffuse and gradually blend with surrounding skin. Commonly affects the lower legs (often associated with tinea pedis as a portal of entry). May be associated with lymphangitis (red streaks extending proximally from the infection site) and regional lymphadenopathy.

Management: Oral flucloxacillin 500 mg QID x 7 days (or clarithromycin 500 mg BD if penicillin-allergic) for mild-to-moderate disease. IV antibiotics (flucloxacillin 1–2 g QID, or benzylpenicillin 1.2 g QID + flucloxacillin 1–2 g QID) for severe disease, facial erysipelas, or systemic sepsis. Elevation of the affected limb is important to reduce oedema.

Folliculitis, Furuncles, and Carbuncles

- **Folliculitis:** Superficial infection of the hair follicle, presenting as erythematous papules or pustules centred on a hair follicle. Commonly caused by *S. aureus*.
- **Furuncle (boil):** Deep infection of the hair follicle extending into the surrounding dermis and subcutaneous tissue, presenting as a tender, erythematous, fluctuant nodule.
- **Carbuncle:** A cluster of interconnected furuncles, presenting as a large, deep, multiloculated abscess with multiple draining sinuses. More common in diabetics and immunocompromised patients.

Management: Small furuncles may resolve with warm compresses. Larger or persistent lesions require incision and drainage. Oral antibiotics (flucloxacillin) are indicated for extensive disease, systemic symptoms, or lesions in high-risk areas (face — risk of cavernous sinus thrombosis).

Fungal Infections (Dermatophytosis Tinea)

Dermatophytes are filamentous fungi that infect keratinised tissues (skin, hair, nails). The three main genera are *Trichophyton*, *Microsporum*, and *Epidermophyton*. Infections are named by the site of involvement:

- **Tinea corporis (ringworm of the body):** Annular (ring-shaped), erythematous, scaly plaques with central clearing and an active, raised, scaly border. Usually unilateral or asymmetrical. Topical antifungals (terbinafine 1% cream BD x 2–4 weeks) are effective for localised disease. Oral terbinafine 250 mg daily x 2–4 weeks for extensive or refractory disease.
- **Tinea pedis (athlete's foot):** The most common dermatophyte infection. Three patterns: interdigital (scaling, maceration, fissuring between the toes — especially the 4th/5th web space), moccasin (diffuse scaling of the soles and lateral feet), and vesiculobullous (vesicles/bullae on the instep and sole). Topical terbinafine 1% cream BD x 1–2 weeks. Oral terbinafine for refractory cases.
- **Tinea capitis (scalp ringworm):** Most common in children. Presents with patchy hair loss (alopecia) with broken hairs ("black dot" pattern in *Trichophyton tonsurans* infection), scaling, and sometimes a kerion (a boggy, inflamed, purulent mass). Oral antifungals are required (griseofulvin 10–20 mg/kg/day x 6–8 weeks, or terbinafine 250 mg daily x 4 weeks for *Trichophyton* species). Topical antifungals alone are insufficient for tinea capitis.
- **Tinea unguium (onychomycosis):** Fungal infection of the nail. Presents with thickened, discoloured (yellow/brown), brittle, dystrophic nails. Diagnosis should be confirmed by nail clipping for mycology (fungal culture and/or PAS stain) before starting treatment, as clinical diagnosis alone is unreliable (only ~50% of dystrophic nails are due to fungal infection). Oral terbinafine 250 mg daily x 6 weeks (fingernails) or 12 weeks (toenails) is first-line. Cure rates are approximately 70–80% for fingernails and 50–70% for toenails.

Viral Infections

Herpes Simplex Virus (HSV)

HSV-1 (typically orolabial) and HSV-2 (typically genital) cause recurrent vesicular eruptions. Primary infection may be severe (e.g., herpetic gingivostomatitis in children, primary genital herpes). Recurrent infections are triggered by stress, UV exposure, immunosuppression, and menstruation.

Management: Aciclovir 400 mg TID x 5 days (or valaciclovir 500 mg BD x 5 days) for episodic treatment. Suppressive therapy (aciclovir 400 mg BD) for frequent recurrences (>6 per year).

Herpes Zoster (Shingles)

Reactivation of latent varicella-zoster virus (VZV) in a dorsal root or cranial nerve ganglion. Presents with a painful, unilateral, dermatomal vesicular eruption. The most common sites are the thoracic dermatomes and the ophthalmic division of the trigeminal nerve (herpes zoster ophthalmicus — requires urgent ophthalmology review due to risk of corneal damage).

Post-herpetic neuralgia (PHN): Persistent pain lasting >90 days after the onset of the rash. Risk increases with age (>50 years). Management includes gabapentin, pregabalin, tricyclic antidepressants (amitriptyline), topical capsaicin, and lidocaine patches.

Management: Antiviral therapy (aciclovir 800 mg 5 times daily x 7 days, or valaciclovir 1 g TID x 7 days) should be started within 72 hours of rash onset to reduce the severity and duration of the acute episode and the risk of PHN. Analgesia (paracetamol, NSAIDs, opioids for severe pain) is important.

Warts (Verrucae)

Warts are caused by human papillomavirus (HPV) and present as hyperkeratotic papules. Types include common warts (verruca vulgaris — hands, fingers), plantar warts (verruca plantaris — soles of feet), plane warts (flat-topped papules — face, dorsum of hands), and genital warts (condylomata acuminata — anogenital region, associated with HPV types 6 and 11).

Management: Many warts resolve spontaneously within 2 years. Treatment options include topical salicylic acid (first-line for common and plantar warts), cryotherapy (liquid nitrogen applied with a cryospray or cotton bud, repeated every 2–4 weeks), and curettage under local anaesthesia. Genital warts may be treated with topical imiquimod, podophyllotoxin, or cryotherapy.

Common Skin Infections — Summary

Infection	Organism	Key Features	First-Line Treatment
Impetigo	S. aureus, S. pyogenes	Honey-coloured crusts; face and limbs	Topical fusidic acid; oral flucloxacillin if extensive
Cellulitis/erysipelas	S. pyogenes, S. aureus	Erythema, warmth, swelling; erysipelas has sharp border	Oral flucloxacillin; IV if severe
Tinea corporis	Dermatophytes	Annular, scaly plaque with central clearing	Topical terbinafine BD x 2–4 weeks
Tinea capitis	Trichophyton, Microsporum	Patchy alopecia, scaling; kerion	Oral griseofulvin or terbinafine
Herpes zoster	VZV	Painful, dermatomal vesicular eruption	Aciclovir 800 mg 5x/day x 7 days (within 72 hours)
Warts	HPV	Hyperkeratotic papules	Topical salicylic acid; cryotherapy

Pigmentary Disorders

Hyperpigmentation

Melasma

Melasma is an acquired hyperpigmentation disorder presenting as symmetrical, brown-grey, macular patches on the face (centrofacial, malar, or mandibular distribution). It is more common in women, particularly during pregnancy ("mask of pregnancy") and in those taking the combined oral contraceptive pill. Sun exposure is a major exacerbating factor.

Management: Rigorous sun protection (broad-spectrum SPF 50+ sunscreen, wide-brimmed hat) is essential. Topical treatments include hydroquinone 2–4% (used in cycles of 3 months on, 3 months off due to risk of exogenous ochronosis), azelaic acid 15–20%, topical retinoids, and combination creams (Kligman's formula: hydroquinone + tretinoin + topical steroid). Chemical peels and laser therapy may be considered in refractory cases.

Post-Inflammatory Hyperpigmentation (PIH)

PIH occurs following inflammation or injury to the skin (e.g., acne, eczema, burns, laser procedures). It is more common and more pronounced in darker skin tones (Fitzpatrick IV–VI). Management includes treatment of the underlying condition, sun protection, topical retinoids, azelaic acid, vitamin C, and chemical peels.

Hypopigmentation

Vitiligo

Vitiligo is an acquired, autoimmune condition characterised by the destruction of melanocytes, resulting in well-demarcated, depigmented (milky-white) macules. It affects approximately 0.5–1% of the population and can occur at any age. Common sites include the face, hands, wrists, body folds, and around body orifices (periorificial). Vitiligo is associated with other autoimmune conditions, including thyroid disease (Hashimoto's thyroiditis, Graves' disease), type 1 diabetes mellitus, pernicious anaemia, and Addison's disease.

Management: Sun protection for depigmented areas (no melanin = no UV protection). Cosmetic camouflage. Topical corticosteroids or calcineurin inhibitors for localised disease. NB-UVB phototherapy for generalised disease. In stable, localised disease, surgical techniques (split-thickness skin grafting, melanocyte-keratinocyte transplantation) may be considered. Janus kinase (JAK) inhibitors (e.g., ruxolitinib cream) have recently shown promise in repigmentation.

Pityriasis Versicolor

Pityriasis versicolor is a superficial fungal infection caused by *Malassezia* yeast. It presents as multiple, well-demarcated, scaly, hypopigmented (or hyperpigmented) macules on the trunk and proximal limbs. The lesions may coalesce into larger patches. A fine scale is visible when the skin is scraped. It is more common in warm, humid climates and in young adults.

Diagnosis: Wood's lamp examination may show golden-yellow fluorescence. Skin scraping with KOH preparation reveals short hyphae and spores ("spaghetti and meatballs" appearance).

Management: Topical ketoconazole 2% shampoo (applied to affected areas, left for 5 minutes, then rinsed off — used daily for 5 days, then weekly for several months to prevent recurrence). Oral fluconazole 300 mg weekly x 2 weeks or itraconazole 200 mg daily x 7 days for extensive or refractory disease. Recurrence is common (up to 60% within 2 years).

Benign and Malignant Skin Lesions

Benign Lesions

Seborrhoeic Keratoses

Seborrhoeic keratoses are extremely common, benign epidermal tumours that increase in number with age. They present as well-demarcated, waxy, "stuck-on" appearing papules or plaques with a rough, verrucous surface. Colour varies from light brown to black. They may be numerous (the "sign of Leser-Trélat" — sudden eruption of multiple seborrhoeic keratoses — may indicate an underlying malignancy, particularly gastric adenocarcinoma). No treatment is required unless they are symptomatic or cosmetically bothersome. Options include curettage, cryotherapy, or shave excision.

Dermatofibroma

Dermatofibroma is a benign dermal fibrous nodule, most common on the legs of young to middle-aged adults. It presents as a firm, dome-shaped, skin-coloured to brown papule or nodule. A characteristic clinical sign is the "dimple sign" — lateral compression causes central dimpling. No treatment is required.

Melanocytic Naevi (Moles)

Melanocytic naevi are benign proliferations of melanocytes. They are classified by their location:

- **Junctional naevus:** Flat, dark brown macule (naevus cells at the dermal-epidermal junction)
- **Compound naevus:** Slightly raised, brown papule (naevus cells in the epidermis and dermis)
- **Intradermal naevus:** Dome-shaped, skin-coloured or light brown papule (naevus cells confined to the dermis)

Most adults have 10–40 melanocytic naevi. They typically appear in childhood and adolescence, darken during pregnancy, and may fade in later life. The vast majority are benign and require no treatment.

Malignant Lesions

Melanoma

Melanoma is a malignant tumour of melanocytes. It is the most serious form of skin cancer and accounts for the majority of skin cancer deaths. Incidence has been increasing worldwide, particularly in fair-skinned populations.

Risk factors: Fair skin (Fitzpatrick skin types I–II), history of sunburn (particularly childhood sunburn), intermittent intense UV exposure, family history of melanoma, >50 melanocytic naevi, atypical mole syndrome (dysplastic naevus syndrome), immunosuppression.

Clinical subtypes:

- **Superficial spreading melanoma (SSM):** The most common subtype (~70%). Presents as an irregularly shaped, pigmented macule or plaque with variegated colour (brown, black, blue, red, white), irregular borders, and asymmetry. Most common site: lower leg in women, back in men.
- **Nodular melanoma (NM):** The second most common subtype (~15–30%). Presents as a rapidly growing, raised, nodular lesion. May be amelanotic (non-pigmented — red or skin-coloured), which can lead to delayed diagnosis. Most aggressive subtype.
- **Lentigo maligna melanoma (LMM):** Arises from lentigo maligna (melanoma in situ on chronically sun-damaged skin). Presents as an irregularly pigmented macule on the face or other sun-exposed areas of elderly patients.
- **Acral lentiginous melanoma (ALM):** The least common subtype in white populations but the most common in darker-skinned individuals. Presents as an irregularly pigmented macule or plaque on the palms, soles, or beneath the nail (subungual melanoma — presents as a longitudinal pigmented band in the nail, with Hutchinson's sign — pigmentation extending to the proximal or lateral nail fold).

ABCDE criteria for melanoma detection:

- **A**symmetry — one half of the lesion does not match the other
- **B**order irregularity — ragged, notched, or blurred edges
- **C**olour change — variegated colour within the lesion
- **D**iameter > 6 mm (though melanomas can be smaller)
- **E**volving — any change in size, shape, colour, or symptoms (itching, bleeding)

Management: Any suspicious pigmented lesion should be referred urgently (2-week wait pathway in the UK) for dermatological assessment. Diagnosis is confirmed by excision biopsy (not incision or punch biopsy, which may miss the worst area). The key prognostic factor is *Breslow thickness* — the depth of invasion measured in millimetres from the granular layer of the epidermis to the deepest point of tumour invasion. Treatment is wide local excision with margins determined by Breslow thickness. Sentinel lymph node biopsy may be offered for tumours ≥1 mm thickness. Advanced melanoma is treated with immunotherapy (anti-PD-1 agents: pembrolizumab, nivolumab; anti-CTLA-4: ipilimumab) and/or targeted therapy (BRAF inhibitors: vemurafenib, dabrafenib; MEK inhibitors: trametinib) for BRAF-mutant tumours.

Basal Cell Carcinoma (BCC)

BCC is the most common type of skin cancer. It arises from basal keratinocytes of the epidermis and is locally invasive but very rarely metastasises (<0.1% of cases). Risk factors include chronic UV exposure, fair skin, and immunosuppression.

Clinical features: The classic presentation is a pearly, translucent papule or nodule with telangiectasia (visible blood vessels) on the surface. The lesion may ulcerate (rodent ulcer — a term historically used for BCC). Common sites are sun-exposed areas: face (particularly the nose, periorbital region, and ears), neck, and trunk.

Subtypes:

- **Nodular BCC:** Most common. Pearly nodule with telangiectasia.
- **Superficial BCC:** Erythematous, scaly plaque, often on the trunk. May resemble eczema or psoriasis.
- **Morphoeic (sclerosing) BCC:** Scar-like, indurated plaque with indistinct borders. More aggressive and more likely to recur after treatment.

Management: Excision with clear margins is the gold standard. Mohs micrographic surgery (a technique in which the tumour is excised layer by layer and examined microscopically during the procedure to ensure complete removal) is preferred for high-risk BCCs (morphoeic subtype, recurrent tumours, or tumours in cosmetically sensitive areas). Other options include curettage and cautery, cryotherapy, topical imiquimod, topical 5-fluorouracil, and photodynamic therapy (PDT).

Squamous Cell Carcinoma (SCC)

SCC is the second most common type of skin cancer. It arises from keratinocytes of the epidermis and has the potential to metastasise (metastatic risk: 2–6% for cutaneous SCC, higher for high-risk tumours).

Risk factors: Chronic UV exposure (cumulative, unlike BCC which is associated with intermittent exposure), immunosuppression (particularly organ transplant recipients — SCC is 65–250 times more common in transplant recipients), chronic wounds (Marjolin's ulcer — SCC arising in a chronic scar or ulcer), HPV infection, and actinic keratoses (premalignant lesions).

Clinical features: Presents as a rapidly growing, firm, erythematous nodule or plaque, often with surface crusting, ulceration, or keratosis. Common sites: sun-exposed areas (face, ears, dorsum of hands, lower legs). Actinic keratoses (AKs) are premalignant lesions that present as rough, scaly, erythematous macules or papules on sun-damaged skin. They have a small but definite risk of progressing to SCC (estimated 0.025–20% per lesion per year).

Management: Excision with clear margins. Mohs surgery for high-risk tumours. Sentinel lymph node biopsy may be considered for high-risk SCCs. Treatment of actinic keratoses includes topical 5-fluorouracil, topical imiquimod, topical diclofenac gel, cryotherapy, curettage, and photodynamic therapy.

KEY TEACHING POINTS

The ABCDE criteria are a useful screening tool for melanoma, but not all melanomas conform to these criteria. Amelanotic melanomas (non-pigmented) may be missed if relying solely on the ABCDE criteria. Any new, changing, or symptomatic pigmented lesion warrants urgent referral.

The "ugly duckling" sign — a mole that looks different from the patient's other moles — is a useful clinical indicator of melanoma, even if the lesion does not meet the ABCDE criteria.

Organ transplant recipients on long-term immunosuppression have a dramatically increased risk of skin cancer (particularly SCC). They require regular dermatological surveillance and rigorous sun protection.

Actinic keratoses are premalignant — they should be treated to prevent progression to SCC. Field-directed therapy (e.g., topical 5-FU, imiquimod, PDT) is preferred when there are multiple AKs in a sun-damaged area.

MCQ TRAPS & PITFALLS

BCC is locally invasive but almost never metastasises — if a question describes a skin cancer that has metastasised, it is unlikely to be BCC.

SCC arising in a chronic wound or scar is called a Marjolin's ulcer — it has a higher metastatic risk than SCC arising in sun-damaged skin.

Subungual melanoma presents as a longitudinal pigmented band in the nail (longitudinal melanonychia) with Hutchinson's sign (pigment extending to the nail fold). It is often misdiagnosed as a subungual haematoma or fungal infection.

The sign of Leser-Trélat (sudden eruption of multiple seborrhoeic keratoses) is a paraneoplastic sign associated with internal malignancy, most commonly gastric adenocarcinoma.

Comparison of Common Skin Cancers

Feature	BCC	SCC	Melanoma
Origin	Basal keratinocytes	Keratinocytes	Melanocytes
Metastasis risk	Very rare (<0.1%)	2–6% (higher in high-risk tumours)	Significant; depends on Breslow thickness
Typical appearance	Pearly nodule with telangiectasia	Crusted/ulcerated nodule on sun-exposed skin	Irregular pigmented lesion (ABCDE)
Key risk factor	Intermittent UV exposure	Cumulative UV exposure; immunosuppression	Intermittent intense UV; fair skin; sunburn
First-line treatment	Excision; Mohs for high-risk	Excision; Mohs for high-risk	Excision; sentinel lymph node biopsy; immunotherapy/targeted therapy for advanced disease

Practice Questions

Q: A 45-year-old woman presents with a 6-month history of a slowly growing, painless, pearly nodule on the left side of her nose. The lesion has a smooth, translucent surface with visible small blood vessels and a central ulcer. She has a history of chronic sun exposure from working outdoors. What is the most likely diagnosis?

A. Squamous cell carcinoma

B. Basal cell carcinoma

C. Melanoma

D. Seborrhoeic keratosis

E. Actinic keratosis

Answer: B

Explanation: The classic presentation of a pearly, translucent nodule with telangiectasia and central ulceration on a sun-exposed area (nose) in a patient with chronic UV exposure is characteristic of basal cell carcinoma (BCC). BCC is the most common skin cancer and is locally invasive but very rarely metastasises. SCC typically presents as a crusted or ulcerated nodule without the pearly, translucent appearance. Melanoma would typically be pigmented with irregular borders. Seborrhoeic keratoses have a "stuck-on" waxy appearance and do not ulcerate. Actinic keratoses are rough, scaly macules/papules, not nodular.

Q: A 16-year-old boy presents with multiple, well-demarcated, depigmented (milky-white) patches on the dorsal aspects of both hands and around his mouth. The lesions are asymptomatic and have been slowly increasing in size over the past year. He also has a history of Hashimoto's thyroiditis. What is the most likely diagnosis?

A. Pityriasis versicolor

B. Pityriasis alba

C. Vitiligo

D. Post-inflammatory hypopigmentation

E. Tinea corporis

Answer: C

Explanation: Vitiligo presents as well-demarcated, depigmented (milky-white) macules, commonly affecting the hands, face (periorificial), and body folds. It is an autoimmune condition and is associated with other autoimmune diseases, including thyroid disease (Hashimoto's thyroiditis, Graves' disease). Pityriasis versicolor causes hypopigmented (not depigmented) macules with fine scale on the trunk. Pityriasis alba causes ill-defined, hypopigmented patches on the face of children, often with mild eczema. Post-inflammatory hypopigmentation follows a preceding inflammatory skin condition. Tinea corporis presents as an annular, scaly plaque with central clearing, not depigmented patches.

Q: A 28-year-old woman presents with a 3-day history of a painful, blistering rash on the left side of her chest, distributed in a band-like pattern from the spine to the sternum. She reports severe burning pain in the area. She had chickenpox as a child. What is the most likely diagnosis?

A. Herpes simplex virus infection

B. Contact dermatitis

C. Herpes zoster (shingles)

D. Cellulitis

E. Erysipelas

Answer: C

Explanation: Herpes zoster (shingles) is caused by reactivation of latent varicella-zoster virus (VZV) in a dorsal root ganglion. It presents with a painful, unilateral, dermatomal vesicular eruption — classically described as a band-like distribution. The patient's history of chickenpox (primary VZV infection) supports the diagnosis. HSV infection typically affects the orolabial or genital region and does not follow a dermatomal distribution. Contact dermatitis is itchy, not typically painful, and does not follow a dermatomal pattern. Cellulitis and erysipelas present as erythematous, oedematous plaques, not vesicular eruptions in a dermatomal distribution.

Q: A 60-year-old man presents with a 4-week history of a rapidly growing, irregularly shaped, darkly pigmented lesion on his back. The lesion is 8 mm in diameter, has variegated colour (brown, black, and red), irregular borders, and is asymmetrical. He has a history of multiple sunburns in childhood. What is the most appropriate next step in management?

A. Reassurance and review in 3 months

B. Topical corticosteroid cream

C. Urgent referral to dermatology (2-week wait)

D. Shave biopsy in primary care

E. Prescribe oral antibiotics

Answer: C

Explanation: The lesion meets multiple ABCDE criteria for melanoma (Asymmetry, Border irregularity, Colour variegation, Diameter >6 mm, Evolving — rapidly growing). The patient also has risk factors (history of sunburns). Any suspicious pigmented lesion should be referred urgently via the 2-week wait pathway for dermatological assessment. Excision biopsy (not shave biopsy) is the diagnostic procedure of choice. Reassurance, topical steroids, and antibiotics are inappropriate for a suspected melanoma.

Q: A 35-year-old woman presents with a 2-month history of symmetrical, brown-grey patches on her cheeks and forehead. She is currently 20 weeks pregnant and reports that the patches appeared shortly after conception. She has no other symptoms. What is the most likely diagnosis?

A. Addison's disease

B. Melasma

C. Post-inflammatory hyperpigmentation

D. Acanthosis nigricans

E. Drug-induced hyperpigmentation

Answer: B

Explanation: Melasma is an acquired hyperpigmentation disorder that commonly affects the face (centrofacial or malar distribution) and is more common in women. It is frequently triggered by pregnancy ("mask of pregnancy") and the combined oral contraceptive pill. Sun exposure exacerbates the condition. Addison's disease causes generalised hyperpigmentation, particularly in sun-exposed areas, skin creases, and buccal mucosa, but would not present as isolated facial patches. Post-inflammatory hyperpigmentation follows a preceding inflammatory condition. Acanthosis nigricans presents as velvety hyperpigmentation in flexural areas (axillae, neck, groin). Drug-induced hyperpigmentation would require a relevant drug history.

Q: A 55-year-old man with a history of renal transplantation (on long-term immunosuppression with tacrolimus and mycophenolate mofetil) presents with a 3-month history of a rapidly growing, firm, crusted nodule on his left ear. The lesion bleeds easily on contact. What is the most likely diagnosis?

A. Basal cell carcinoma

B. Squamous cell carcinoma

C. Seborrhoeic keratosis

D. Actinic keratosis

E. Keratoacanthoma

Answer: B

Explanation: Squamous cell carcinoma (SCC) is the most common skin cancer in organ transplant recipients on long-term immunosuppression (6